Practice Quiz: From Idea to Intervention

Part A: Multiple Choice

1. In Active Inference, a prototype is best understood as: A) A finished product ready for sale B) A hypothesis expressed in physical form — an action designed to test the inventor's generative model against reality C) A decorative model used only for presentations D) An exact replica of the final design at a smaller scale

2. The Wright Brothers built a small wind tunnel before building a full aircraft. This is an example of: A) Wasting time on unnecessary experiments B) A minimum viable prototype that tested their most critical assumption (the accuracy of published aerodynamic data) with minimal cost C) Copying the testing methods of their competitors D) A pragmatic prototype designed to demonstrate the final product

3. Expected free energy in the context of prototyping balances: A) Cost and revenue B) Epistemic value (reducing uncertainty through learning) and pragmatic value (achieving the goal) C) Material weight and structural strength D) Team size and development speed

4. When a prototype fails, an Active Inference perspective treats this as: A) A waste of time and materials B) Proof that the inventor is incompetent C) A source of specific prediction errors that update the generative model — informative data D) A reason to abandon the project

5. The perception-action loop in prototyping involves: A) Thinking about the prototype without building it B) Building once and accepting the result without modification C) Acting on materials, observing the response, updating the model, and acting again in a continuous cycle D) Asking others to build the prototype while you watch

6. An inventor should transition from epistemic prototyping (learning) to pragmatic prototyping (demonstrating) when: A) They run out of materials B) The critical uncertainties in the generative model have been sufficiently resolved through testing C) They have been working for exactly six months D) Their first prototype works perfectly

7. Chuck Hull's first 3D-printed cup was crude and flawed, but it was valuable because: A) It was the first product he could sell B) It proved the core concept (layer-by-layer UV curing can produce 3D objects) and generated specific prediction errors about layer bonding, shrinkage, and finish that guided further development C) It was aesthetically beautiful D) It required no further iteration

Part B: Short Answer and Analysis

1. An inventor is developing a new type of ergonomic kitchen knife. They are uncertain about two things: (a) whether the blade geometry will cut effectively, and (b) whether users will find the handle comfortable. Design two minimum viable prototypes — one for each uncertainty. For each, describe what you would build, what materials you would use, what you would test, and what a confirming vs. disconfirming result would look like. Which prototype should be built first, and why?

2. James Dyson built 5,127 prototypes over five years. Using the Active Inference framework, explain why such extensive iteration was necessary rather than wasteful. Your answer should address: (a) what each prototype was testing, (b) how prediction errors from each iteration refined the generative model, and (c) why mental simulation alone could not have replaced the physical prototyping process for this invention.

3. You have built a prototype of your invention, and the results are ambiguous — the mechanism works partially but with unexpected side effects. Using the Active Inference framework, describe the three steps you should take next: (a) isolating the specific prediction errors, (b) determining whether your model needs parameter updates or structural changes, and (c) designing the next prototype to resolve the remaining ambiguity. Provide a concrete example from any invention domain.